

A Neural Model for Automatic Bidding of Contract Bridge

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Abstract—Contract bridge is a typical representative of imperfect information game, which consists of two phases: bidding and playing. One of the most challenging tasks in the contract bridge is bidding. Prior studies mainly focus on bidding without competition. However, it is not the case in reality, which inevitably leads to information loss. This work introduces hand understanding to model the real bidding process and restore the multi-task nature of the work. Therefore, Automatic Bidding Model(ABM), a novel neural architecture, is designed to solve this multi-task learning problem. A layered LSTM is provided to players to model real data, as well as an understanding and a bidding component. ABM proves its advantage through comprehensive experiments on real data set. Lastly, this study summarizes the advantage of ABM.

Index Terms—Contract Bridge, Bidding, LSTM, neural network

I. INTRODUCTION

Computational game playing is an emerging topic in artificial intelligence(AI), which is drawing ever-increasing attention from researchers. A large number of researchers focus on perfect information gaming especially with two-player such as chess [1] or go [2], [3] and have been making breakthrough achievements. However, the problem of imperfect gaming in general still remains open. The challenge lies in the information imperfection nature of such gaming, and the information asymmetry renders this problem much tougher.

A few studies investigate imperfect gaming such as Poker [4] or Stratego and consider the approximate solution or improve the current result by solving the disjoint subgames since the precise combination of independent subgames is intractable. This paper focuses on the game of *Contract Bridge*, a typical representative of a multi-player imperfect information game with opposing teams.

Contract bridge, or simply bridge, is a trick-taking card game played by four players, using a standard 52-card deck.

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Four players (North, South, East, and West) are organized into two opposing teams, the North and South against the East and West. A bridge deal consists of two phases, bidding and playing. During the bidding phase, since only the in-hand cards are available to each player, each partnership needs to use an established bidding system to communicate information and obstruct the exchange of information from the competitors. Each partnership will explain the meanings of their actions during the bidding to the opponent when the request is issued. Each bid from the legitimated sequence of 1♣, 1♦, 1♥, 1♠, 1NT, 2♣, ..., 7NT with three flexible bids of DOUBLE, REDOUBLE and PASS suggests a particular trump suit (or there is not a trump suit at all). The dealer initiates the bidding process that ends if a legitimate bid is followed by three sequential PASS calls. The last non-PASS bid is to be the contract of bridge. The player who bids the last non-PASS bid (referred as the contract of the bridge) is called the declarer. The contract affects score that the declarer's team get in the playing phase, which consists of a number and either a suit or no-trump. If the declarer's team achieves the contract, it gets a positive score and the opposing team gets an equal but negative score.

During the playing phase, the left of the declarer leads for the first card, and then every other player, in clockwise order, plays a card of the same suit unless they don't have, in which case they may show any card, particularly the trump suit having some priority. The player with the highest-ranked card wins the round. After the 13 rounds, the score depends on the final contract and the number of the winning rounds of the declarer's team. In order to effectively reduce the lucky factor of strengths of hands in random deals, the scoring method commonly used in international bridge competitions is International Match Points (IMPs), which is an integer between 0 and 24 obtained by the two pairs.

In real human bridge games, people usually think expert human bridge players have similar levels of professional competence in the playing phase. Thus, the win or loss largely

depends on the bidding phase. Considering this practical situation, it is reasonable to study computer bridge bidding. Several algorithmic approaches have been proposed to improve the performance of computer bridge programs in the bidding phase. However, most of these studies have simplified the model to the bidding without competition [5]–[7], do not match the real scene of the bridge, which may lead to information loss.

Hence, this study focuses on the problem of bidding with the competition, which means that each AI player may bid and exchange the information as in the real bridge games. The main contributions of our work are summarized as following:

- Nearly 700,000 instances of real bidding were generated by Synrey platform which has recently won the runner-up in the World Computer Bridge Championship for three consecutive years to train the neural automatic bidding model. This model is the first to use an auxiliary system to assist in making bidding decision bridge bidding, which shows a new method that attempts to solve imperfect information games.
- LSTMs were utilized for the first time in bridge bidding. Instead of having a single layer, a multi-layer LSTM network was trained to process the historical bidding sequence.
- We conduct extensive experiments on the real-world data set and demonstrate the evident superiority.

The rest of this paper is structured as follows. Section II discusses related work. Section III proposes the model framework and describes the algorithms for each phase in the framework. Then we introduce experimental setups and present the results with evaluation in Section IV. Finally, we conclude our work in Section V.

II. RELATED WORK

A. Imperfect Information Games

At present, a large number of current research is based on perfect information games, such as chess [1] and Go [2], [3]. In recent years, with the development of deep learning, the research of perfect information games has made rapid progress. AI players defeated world-class human players, which is an important milestone in the history of artificial intelligence development. In October 2015, AlphaGo won the European Go Champion Fan Wei in 5:0 who is a professional 2 dan player. In March 2016, he challenged the World Go Champion Li Shishi who is the professional 9 dan player to succeed. In October 2017, after making a huge breakthrough in Go, DeepMind has explored other perfect information games, including Atari, chess and Shogi [8]. The paper showed that AlphaGo Zero taught itself three different board games in three days. This result shocked the chess world, and AlphaGo Zero became the best chess player in the world.

However, the above are some researches of perfect information games. There are still many challenges in the field of games with imperfect information due to their partial observability. One of the games that has attracted considerable

attention is poker and most previous work on poker has been limited to Texas Hold'em. The Computer Poker Research Group at the University of Alberta is a typical representative that pays more attention to Texas Hold'em Poker [9]. They have solved heads-up Limit Texas Hold'em in 2015 [10] and heads-up No-Limit Texas Hold'em in 2017 [11]. These systems treat poker as imperfect information games and find specific search strategies for Texas Hold'em to get the game equilibrium. One of the standard approaches to computing strategies in such large games is to first generate many abstractions by combining similar game states together, which is a smaller version of the game that retains as much as possible the strategic characteristics of the original game [12], [13]. Although DeepStack [11] and Libratus [14] have made great progress in heads-up No-Limit Texas Hold'em, multi-player No-Limit Texas is still a technical problem due to the huge size of the space of hidden information. In bridge, each player has 13 cards, therefore there would have been $C_{39}^{13} = 8,122,425,444$ possible states, which is impossible to get the full information. This is why most previous works on poker are limited to two-player Texas Hold'em. As for this problem, we decide to add a hand understanding system which can predict the other players' hand distribution so that transforming imperfect information into perfect information as much as possible.

B. bridge bidding

Bridge is not only an imperfect information game, but also a multi-player game. Bridge has begun to get widespread attention. There are some bridge softwares emerging such as Bridge Baron, GIB [15] and Wbridge, which have proved to be especially effective during the playing phase. Bidding, however, has shown to be more complex.

Therefore, this paper mainly focuses on bridge bidding. One of the first bridge bidding programs was written as early as 1970 [16]. Wassermann [16] introduced bidding patterns. In addition to the opening and responding patterns, this program could distinguish several specific bids, such as double. But this program needs transfer bridge expert's bidding techniques, commonly referred to as human bidding system to program code, which is finite. However, the rules of bridge are extremely complex, which caused the inevitable ambiguity of the bids. To reduce this ambiguity, some works begin by considering human bidding systems. For example, GIB [17] has strengthened the lookahead search in bidding by borrowing rules from human bidding systems. Moreover, Amit et al. [18] attempt to use Monte-Carlo sampling in combination with the partition search to reduce the size of the game tree.

All of the above work is based on the human bidding system. Nevertheless, Ho and Lin [5] proposed a new method adopting a contextual bandit algorithm to build decision trees to predict bids without considering human bidding systems. However, their model is limited to bid from up to 5 options due to the fact that the decision tree has only 5 simple nodes. Yeh and Lin [6] first tried using deep reinforcement

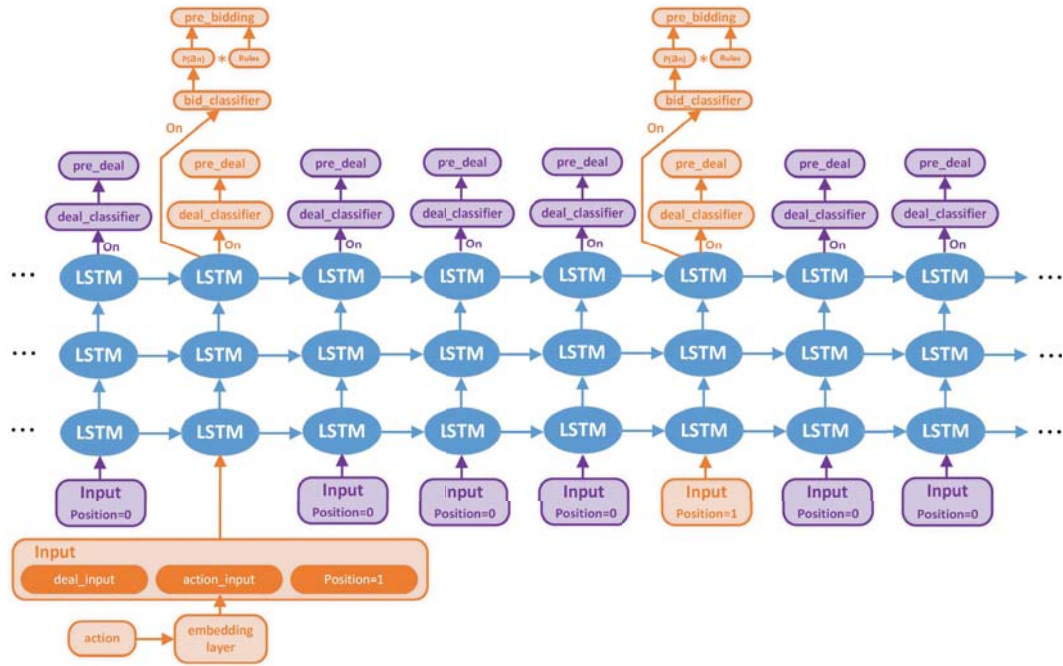


Fig. 1. The model framework for neural bridge bidding model.

learning to bid without depending on any human expert knowledge. They have achieved some promising results, but their successes all rely on simplifying models to the subproblem of bidding without competition which assumes that the opponent's team always calls PASS during the bidding. This situation is extremely rare in real bridge games, which leads that their model is less functional. In contrast, DeLooze and Downey [19] focus on the problem of bidding with the competition. They use raw data as input for a special form of a neural network, called a self-organizing map (SOM). Unfortunately, this approach is only used to bid no trump hands rather than any bids. But Rong et al. [20] considered the problem of bid with competition. We borrow some ideas from [19]. This work generated a large number of examples from Synrey platform, and some of these examples as the input of our model. We also extend some ideas from [7]. For example, a hand understanding system was added to help complete the automatic bidding strategy.

C. bidding system

Bridge bidding is based on a bidding system, which has a set of bidding rules that cover different situations and two opposing teams agree on. There are many different bidding systems in the world. The bidding system is divided into the natural bidding system and the non-natural bidding system. A well-known natural bidding system is Two Over One(2/1), which is used widely by both amateur and professional players. This model used real data from Synrey platform with a bidding system called CCBA that is defined by China Contract Bridge Association. The CCBA is roughly the same as

Two Over One(2/1), with slightly different details. Therefore, the bidding system trained by the proposed automatic bidding model (ABM) is based on CCBA.

Generally, beginners can only memorize the bidding system, but the bidding system itself is a tool to help players achieve goal exploration and goal achievement. The bidding system used by masters is usually more delicate and complex, allowing players to better explore and confront. But in essence, the bidding process is mastered by players, and the bidding system is just a protocol or language to convey information.

III. MODEL FRAMEWORK

This study proposes a multi-task neural learning model for bridge bidding with the competition, referred to as ABM, illustrated in Fig. 1. A fundamental problem in imperfect information games is that states can not be accurately predicted. In order to solve this problem, combining with the actual situation of the bridge, we firstly added a hand understanding system to assist in making a bidding decision. This turns the original problem into a multi-task learning problem [21], [22]. During the learning process, this model not only predicts the next bid but also the hands for four players. Briefly, the model uses the double dummy data provided by the Synrey platform and is trained by a deep neural network to design an automatic bidding machine based on expert experience.

In the traditional deep neural network, it only pays attention to the information at the current moment, rather than what information at the last moment can be used for the next moment. However, for the bridge bidding process, the current

bidding information cannot be regarded as the state. The current historical bidding sequence directly affects the next bid. Recurrent Neural Network (RNN) shows its advantages on this sequence prediction. But, training traditional RNN could encounter the exploding and vanishing gradient problems when dealing with long-term series. LSTM as a special RNN has solved these problems very well. Therefore, we design an automatic bidding model based on a LSTM network.

A common LSTM cell consists of an input gate, an output gate and a forget gate, which keeps the features of the input. The structure of the LSTM unfolded into a full neural network.

$$\begin{aligned}
\mathbf{f}_n &= \sigma(\mathbf{W}_f[\mathbf{h}_{n-1}, \mathbf{x}_n] + \mathbf{b}_f) \\
\mathbf{i}_n &= \sigma(\mathbf{W}_i[\mathbf{h}_{n-1}, \mathbf{x}_n] + \mathbf{b}_i) \\
\tilde{\mathbf{c}}_n &= \tanh(\mathbf{W}_c[\mathbf{h}_{n-1}, \mathbf{x}_n] + \mathbf{b}_c) \\
\mathbf{c}_n &= \mathbf{f}_n \circ \mathbf{c}_{n-1} + \mathbf{i}_n \circ \tilde{\mathbf{c}}_n \\
\mathbf{o}_n &= \sigma(\mathbf{W}_o[\mathbf{h}_{n-1}, \mathbf{x}_n] + \mathbf{b}_o) \\
\mathbf{h}_n &= \mathbf{o}_n \circ \tanh(\mathbf{c}_n)
\end{aligned} \tag{1}$$

where, $\mathbf{W}_f$, $\mathbf{W}_i$, $\mathbf{W}_c$ are the weight matrices, $\mathbf{b}_f$, $\mathbf{b}_i$, $\mathbf{b}_c$ and $\mathbf{b}_o$ are the bias vector parameters. $\mathbf{f}_t$ determines what information can pass through the cell state. $\mathbf{i}_t$ and $\mathbf{c}_t$ determine the newly added information to the cell state. The $\circ$ denotes Hadamard product.

The LSTM contains the input $\{\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}_{n+1}, \dots\}$ and the output $\{\mathbf{h}_1, \dots, \mathbf{h}_n, \mathbf{h}_{n+1}, \dots\}$, where $\mathbf{x}_n$ represents the input in step n and $\mathbf{h}_n$ represents the input in step n . We found that the only useful information to predict the next bidding is the distribution of player's hands and the current historical bidding sequence when setting up the initial training vector of the LSTM network. So we define $\mathbf{x}_n$ containing $\mathbf{D}_i$ and $\mathbf{a}_n$, where $\mathbf{D}_i$ denotes the hands' distribution of Player i ($i = 1, 2, 3, 4$), and $\mathbf{a}_n$ denotes the bid called in the n rounds, which is a 256-dimensional vector according to the embedding layer. $\mathbf{D} = [\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_3, \mathbf{D}_4]^T \in \mathbb{R}^{4 \times 52}$ and $\mathbf{D}_{:j}$ is the j -th column of $\mathbf{D}$.

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As for LSTM representation, we update the parameters by training subtasks. The output of the LSTM $\mathbf{h}_n$ is used as input to two classifiers that can show the distribution of hands and then predict the next bidding.

On the one hand, in the hand standing system that consists of two 512-dimensional fully connected layers, we use cross-entropy cost as loss function $\mathcal{J}_{deal}$, which can be solved as a joint minimization problem:

$$\begin{aligned}
\mathcal{J}_{deal} &= \sum_{n=1}^N \sum_{j=1}^{52} \mathcal{H}(\mathbf{D}_{:j}, {}^n\hat{\mathbf{D}}_{:j}), \\
{}^n\hat{\mathbf{D}} &= \eta(\theta_\eta, h_n),
\end{aligned} \tag{2}$$

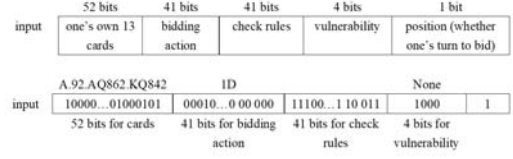


Fig. 2. An example of model input.

where ${}^n\hat{(\cdot)}$ denotes the predicted value in the n -th iteration. $\theta_\eta = \{\mathbf{W}_\eta, \mathbf{b}_\eta\}$ denotes the parameters of the η fc layer, and $\mathcal{H}(\cdot, \cdot)$ is the cross-entropy.

On the other hand, the bidding system consists of two 512-dimensional fully connected layers. Because of the huge difference in the number of pass data and non-pass data, we do not continue to choose cross-entropy as a loss function for this type of imbalance problem, but instead adopt a new loss function: focal loss [23], which can reduce the weight of easily categorized samples and make the model more focus on difficult samples during the training. The loss function $\mathcal{J}_{action}$ can be solved as a joint minimization problem:

$$\begin{aligned}
\mathcal{J}_{action} &= \sum_{k=0}^{4k+i \leq N} FL[P(a_{4k+i}), P(\hat{a}_{4k+i})], \\
P(\hat{a}_{4k+i}) &= \text{softmax}[\tilde{P}(a_{4k+i})], \\
\tilde{P}(a_{4k+i}) &= \pi(\theta_\pi, h_{4k+i}) \odot P_r(a_{4k+i}), \\
i &= 1, 2, 3, 4.
\end{aligned} \tag{3}$$

where $FL[\cdot, \cdot]$ is the focal-loss, $P(a_n)$ is the probability of the event a_n occurring, and θ_π denotes the parameters of the π fc layer. Because our model uses multi-task neural network learning, we need to share the parameters of the LSTM. Finally, we obtain the overall loss as follows:

$$\min \quad \mathcal{J} = \mathcal{J}_{deal} + \mathcal{J}_{action}. \tag{4}$$

Figure2 and Figure3 respectively show an example of model input and output. We can see that the input of the model is the player's hands, bidding action, check rules, vulnerability and position, followed by 3-layers LSTM. A 52-dimensional vector is used to represent the player's hands and a 41-dimensional vector is used to check rules which can ensure the rationality of the bidding sequence. We also add the vulnerability to the input for better score calculation. Besides, one of the most important factors is the position, which means whether one's turn to bid. The importance of this factor will be demonstrated in the experiment later. In Figure3, the output layer of our model consists of hand understanding system and bidding decision making.

IV. EXPERIMENTAL EVALUATION

Before presenting the experimental results, we introduce the data set, network training and comparison methods in each setting. After that, we make some analysis and discussion.

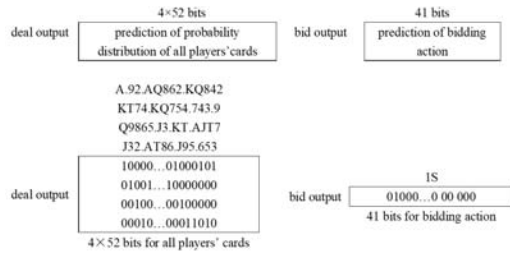


Fig. 3. An example of model output.

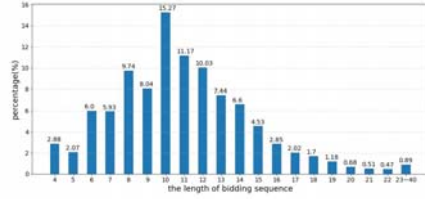


Fig. 4. The distribution of lengths of bidding sequences.

A. Data Description

The data set in the experiment is provided by the Snyrey platform. The Snyrey platform has accumulated nearly 500 million historical records of bidding and playing with the result of double dummy hands. We randomly generated nearly 700,000 instances of real bidding as our data set, 70% of which are used as training data for our automatic bidding model, 20% of which as a validation set, and the remaining data as testing data. Each instance contains four real hands, historical bidding sequence and double-dummy hands results. The distribution of lengths of bidding sequences in the data set are depicted in Figure 4, which basically presents a normal distribution. Most of the bids are concentrated at the end of 8~13, and the probability of too long or short bid sequences is small.

B. Network Training

This model is based on deep neural network. In the embedding layer, the bidding action was converted into a vector with a fixed size, which combining the 52-dimensional hand information distribution will be the state input of the model. At the same time, the hand distribution and the factor

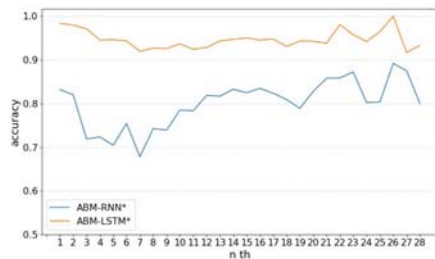


Fig. 5. The accuracy of ABM-RNN* and ABM-LSTM*.

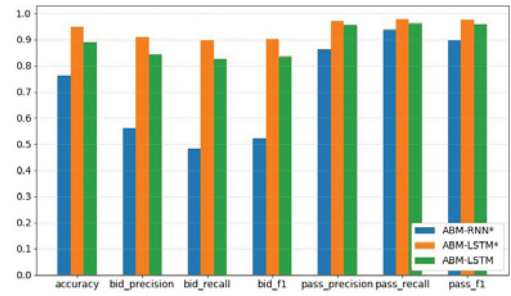


Fig. 8. The performance comparison of different models.

of action's position are mutually constrained in order to train our model accurately. The prediction results will be as output to feed the fully connected layer of the hand understanding system and bidding decision making. All of the models are run on a single GPU, with roughly a runtime of 20 minutes per epoch for model1. We run all the models up to 30 epochs.

First of all, RNN and LSTM network structures were tested. As can be seen from Fig 5, there is a large gap between the model using RNN network structure and the model of LSTM network structure. For each bidding prediction, the accuracy of ABM-LSTM* is much higher than ABM-RNN*.

As for LSTM representation, three main hyper-parameter selections have been made using cross validation technique based on the kl-means and focal-loss. This includes learning rates (constant =0.001 or decreased according to linear or exponential schedule), hidden layer dimensions, number of hidden layers. Figure 6 and 7 shows that we explore the number of hidden layers in [1, 2, 3] and the hidden state size in [128, 256, 512, 1024]. We evaluate the model on validation sets by precision, recall, f1-score and accuracy to choose the best parameters. Finally, we use a 3-layers LSTM and choose 512 as hidden state size in our experiments.

C. Performance of the Model

Next, to show the importance of the position, we trained another neural network without consideration of the factor of the position, which has the same structure with ABM-LSTM*. To get a better insight into the impact of the factor, we compare ABM-LSTM(without the position) with ABM-LSTM*(with the position) on testing sets under different evaluation indexes. It is not difficult to see that adding this feature greatly enhances the experimental effect in Fig 8 show more excellent performance under every evaluation index. To sum up, the components in our models all make contributions to prediction.

D. Comparison with the State-of-the-Art

We compare the bridge bidding result with that in the work of [6], which is a well-known bridge bidding model using deep reinforcement learning. We run the same 140,000 data on our model and that proposed [6]. We choose their deep reinforcement learning model with layer=4 which have

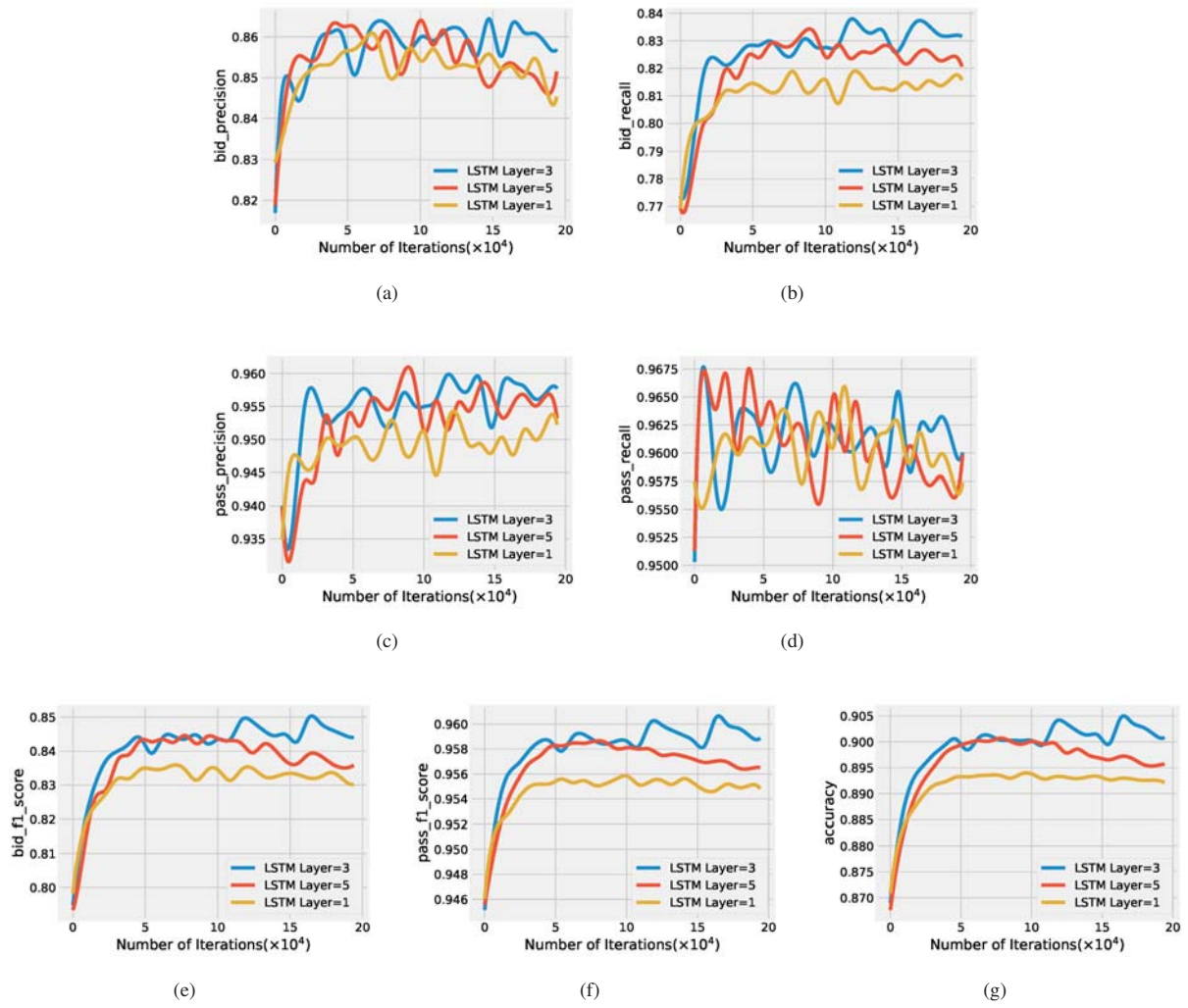


Fig. 6. The evaluation of hyper parameter:layer

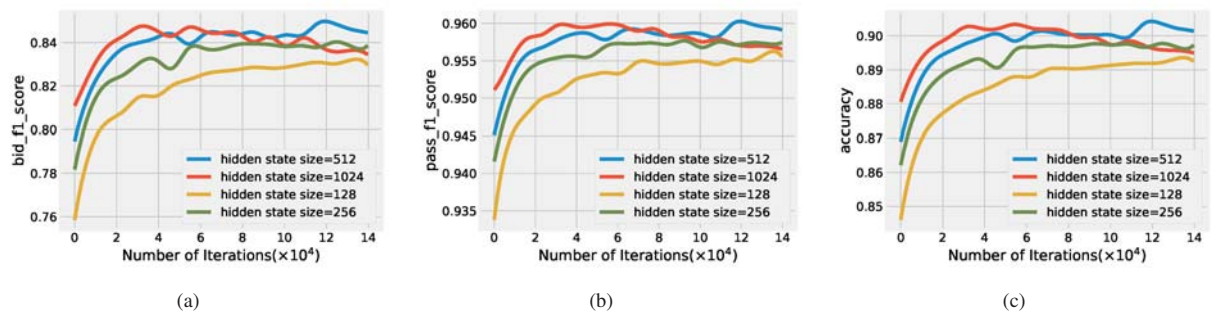


Fig. 7. The evaluation of hyper parameter: hidden state size

TABLE I
COMPARE THE AVERAGE DIFFERENCE OF IMP BETWEEN OUR MODEL
AND VARIOUS DIFFERENT MODELS IN [6]

	our	layer=2	layer=3	layer=4	layer=5
20K	-0.203	-2.143	-2.690	-5.254	-6.130
140K	-0.167	-2.096	-2.658	-5.206	-6.116
200K	-0.178	-2.086	-2.664	-5.244	-6.145

the best performance among all models they think. Because they study the problem of bidding without competition, we provide a method based on the difference between the best contracted score and the actual contracted score, which can be used to evaluate the quality of the bid. The result in Table I demonstrates that our model has the best performance among all models, although this model is simpler without using reinforcement learning.

V. CONCLUSION AND FUTURE WORK

In this paper, we first defined the challenging problem of bridge bidding with competition and proposed a novel multi-task neural learning model based LSTM network to learn to bid automatically. Our ABM was trained with nearly 700,000 instances of real bidding provided by Synrey platform. To the best of our knowledge, our designed model is the first to use an auxiliary system to assist in making bidding decision bridge bidding, which shows a new method that can attempt to solve imperfect information games. We compare our model with deep reinforcement learning models proposed in [6]. Although the structure of our model is more simple, our model can get higher scores in the same game. The result justifies that there is indeed a considerable potential to learn to bid in a data-driven manner. At the same time, our model brings the possibility of tree search for imperfect information games. In 10,000 games, using our model for tree search to assist Synrey platform to play bridge has won 1000 IMPs more than the original software which has won the runner-up in the World Computer Bridge Championship for several years. Thus, we need to improve and continue this work in the future. We will visualize the information of situation understanding for direct communication between players. At the same time, we will combine Monte Carlo sampling and search algorithms for efficient sampling in order to produce higher quality bidding decisions. Besides, use reinforcement learning to further improve the accuracy of the bidding, so that the model gradually forms a reasonable bidding system in the self-game [24].

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